

Zhenxiao Qi

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Research Objective

Zhenxiao Qi is a fifth-year Ph.D. candidate. His research focuses on advancing program analysis and threat detection. He developed various program analysis techniques (e.g., dynamic emulation, logic inference, etc.) to solve security problems, including vulnerability detection, kernel rootkit detection, memory forensics, etc.

Education Background

- ✧ **University of California, Riverside (UCR)** 08/2018 - Present
 - Degree expected: Ph.D. Computer Science.
- ✧ **University of California, Riverside (UCR)**
 - Degree: Master of Science, Computer Science.
- ✧ **University of California, Riverside (UCR)** 07/2017 - 06/2018
 - Exchange student, Computer Science.
- ✧ **Xidian University (XDU)** 07/2014 - 06/2017
 - Degree: Bachelor of Engineering, Information Security

Working Experience

- ✧ **Summer research intern @ Google** (Host: Mithun Iyer) 06/2022 – 09/2022
 - Detection of malicious activities and threats on Google Cloud Platform VMs.
 - Speed up locating kernel memory from large memory snapshots.
- ✧ **Software engineering intern @ Deepbits** (Host: Xunchao Hu) 10/2021 – 06/2022
 - Develop kernel rootkit and threat detection services to monitor and protect cloud VMs and containers.
- ✧ **Summer research intern @ Google** (Host: Mahesh Pisal) 06/2021 - 09/2021
 - Develop and deploy a logic inference-based profile generation approach for kernel memory analysis on Google cloud VMs.
- ✧ **Graduate student researcher @ UCR** (Advisor: Prof. Heng Yin) 08/2018 - Present
- ✧ **Teaching assistant @ UCR** 09/2019 - 03/2021
 - CS202 Advanced Operating System: Fall '19 / Winter '20
 - CS153 Design of Operating systems: Fall '20 / Fall '22 / Spring '22
 - CS100 Software Construction: Spring '20 / Winter '21

Publications:

- [NDSS'22] Zhenxiao Qi, Yu Qu, Heng Yin, [LogicMem: Automatic Profile Generation for Binary-Only Memory Forensics via Logic Inference](#), to appear in the Network and Distributed System Security Symposium, February 2022.
- [NDSS'21] Zhenxiao Qi, Qian Feng, Yueqiang Cheng, Mengjia Yan, Peng Li, Heng Yin, and Tao Wei, SpecTaint: [SpecTaint: Speculative Taint Analysis for Discovering Spectre Gadgets](#), to appear in the Network and Distributed System Security Symposium, February 2021.
- [RAID'19] Ali Davanian, Zhenxiao Qi, Yu Qu, and Heng Yin, [DECAF++: Elastic Whole-System Dynamic Taint Analysis](#), in the 22nd International Symposium on Research in Attacks, Intrusions and Defenses, September 2019.

Research Projects

Speed up locating kernel memory among large memory snapshots.

06/2021 - 09/2022

Mentor: Mithun Iyer

- ▣ VMTD computes kernel code hashes to detect any malicious/abnormal code running in kernel.
- ▣ Locating kernel memory was done via sequential searching which takes minutes when memory snapshot is large.
- ▣ Developed a new method that walks the page table and filters out user-level pages based on PTE flags.
- ▣ Reduced the searching time from minutes to 1 second.

Developed agent-less dynamic container monitoring techniques.

01/2022 – 05/2022

Mentor: Xunchao Hu

- ▣ Developed a dynamic container monitor approach without running any agent inside the container.
- ▣ Automatically collect activities of running processes inside the container, such as network activities, opened files, process credentials and permissions, etc.
- ▣ Performed large scale analysis of existing docker images hosted on DockerHub.

Developed memory snapshot analysis techniques to detect hidden kernel rootkit and attacks in cloud virtual machines

10/2021 - 01/2022

Mentor: Xunchao Hu

- ▣ Developed API-based services to automatically analyze given memory snapshot.
- ▣ Worked on new techniques to discover hidden kernel rootkit and threats based on information obtained from analyzing the memory snapshot, including running process information, network activities, opened files, process credentials and permissions, etc.

Developing and deploying a profile generation approach for kernel integrity analysis on Google Cloud VMs

06/2021 - 09/2021

Mentor: Mahesh Pisal

- ▣ Worked with the Virtual Machine Threat Detection team to improve the coverage of kernel threat detection.
- ▣ Around 20% of VMs are not supported because of lacking workable profiles for their kernel.
- ▣ Integrated the research prototype LogicMEM with VMTD' threat detection tool to enable the support of uncovered VMs.

LogicMEM: A Logic Inference Approach to Automate Profile Generation for Binary-only Memory Forensics

04/2020 - 06/2021

Advisor: Prof. Heng Yin

- ▣ The first binary-only approach that generates kernel profiles for downstream memory forensics.
- ▣ Performed source code kernel evolution analysis and summarized kernel object invariants.
- ▣ Widely tested on a range of Linux kernels from official distributions and showed high precision and recall.

SpecTaint: Speculative Taint Analysis for Discovering Spectre Gadgets

07/2019 - 12/2020

Advisor: Prof. Heng Yin

- ▣ A novel Spectre V1 gadget detection approach by enabling dynamic taint analysis on speculative execution paths.
- ▣ The evaluation results show significantly improved precision and recall with reasonable efficiency.
- ▣ Detected new Spectre V1 gadgets from real-world programs such as Brotli and Caffee.

QEMU Virtualization (KVM)/Emulation (TCG) Mode Switch

04/2018 - 06/2018

Advisor: Prof. Heng Yin

- ▣ Achieved on-the-fly KVM/TCG mode switching on QEMU.
- ▣ Enabled dynamic analysis that benefits from native speed of KVM when instrumentation is not needed.
- ▣ Evaluated on common-used operating systems, such as Windows, Debian, Ubuntu, etc.

DECAF upgrade: porting to the latest version of QEMU

06/2018 - 09/2018

Advisor: Prof. Heng Yin

- ▣ DECAF is a dynamic binary analysis platform based on QEMU.
- ▣ Upgraded DECAF to the latest version of QEMU, and ported the functionalities of DECAF, such as Virtual Machine Introspection, taint analysis, etc.

Professional Services:

Reviewer: [Cybersecurity](#), [Computers and Security](#), [Journal of Cyber Security Technology](#).

Sub-reviewer: USENIX Security'21, DIMVA'19

Artifact Evaluation:

- ▣ ACSAC'18 TIFF: Using Input Type Inference to Improve Fuzzing
- ▣ ACSAC'19 Speculator: A Tool to Analyze Speculative Execution Attacks and Mitigations

Honors and Awards

- Outstanding Teaching Assistance in CSE Department 2019-2020
- Dean's Distinguished Fellowship, UC, Riverside 2018
- Outstanding Student Scholarship, Xidian University 2014-2017

Programming Skills:

Programming Language:

- Fluent in **C**, **Python**, **LaTeX**.
- Other language: **C++**, Java assembly language, MATLAB.

Operating Systems:

- **Linux**, Windows, XV6.

Tools and Platforms:

- AFL, Angr.
- QEMU, DECAF.
- TensorFlow, Caffe.
- Hadoop, Spark.